Brazil
Administrative Council for Economic Defense (CADE)

I. The Context

The Brazilian Administrative Council for Economic Defense (Cade) is advancing its Cerebro project aimed at combating cartels in public procurement, by utilizing Brazil's transparent data ecosystem. Platforms such as Comprasnet, established in 1997, and the Portal Nacional de Compras Públicas (PNCP), strengthened by the 2021 public procurement law, provide detailed data on tenders, bidders, and contracts. These resources enable Cerebro to develop data-driven tools for detecting bid-rigging and supporting investigations, reinforcing Cade's mission to protect economic competition.

Leveraging the latest developments of the Cerebro project, the Novo Rumo operation, launched in December 2024, marked a milestone in the fight against cartels and corruption. The operation targeted a R$9 billion highway infrastructure cartel. Cade conducted this diligence with the cooperation of Controladoria-Geral da União (CGU) and the Polícia Rodoviária Federal (PRF). Initiated in response to a Tribunal de Contas da União (TCU) report, it analyzed tenders from the two largest contractors of such services within the Brazilian federal public administration, between 2018 and 2023, using advanced analytics to identify 12 suspect firms. Enabled by a 2023 Cade-CGU agreement, the operation highlights Cerebro's expanding role in data-driven enforcement.

This achievement outlines Cerebro's tools and challenges, emphasizing its ability to transform procurement data into actionable insights.

II. Techniques and Computational Tools

The Cerebro project transforms Brazil's extensive public procurement data into powerful tools for detecting cartels, enabling investigators to identify illicit activities and respond swiftly to public complaints. Its strength lies in combining robust data analysis with storytelling, crafting automated, reproducible reports that make complex economic evidence accessible to both business units and non-technical audiences, such as judges, ensuring clarity in legal proceedings, as demonstrated in Novo Rumo's evidence synthesis. By integrating traditional and AI-driven methodologies, Cerebro project delivers insights that bridge technical precision with practical applicability, enhancing efficiency across cases.

The project employs a dual-approach toolkit that systematically organizes data into firm-centric categories (such as IP addresses and corporate structures) and tender-centric categories (such as bidding patterns and 'rabbit' firms) to construct robust evidence of collusive behavior. The success of Novo Rumo investigation, underpinned by these methodologies, demonstrates Cerebro's ability to customize outputs for a diverse array of stakeholders. Current initiatives focus on enhancing automation and natural language processing (NLP) capabilities to optimize further streamline analytical processes and improve reproducibility, thereby ensuring consistent and reliable results over time.

From Novo Rumo's experience we can highlight the following:

- The utilization of tools based on data from Comprasnet and PNCP to detect bid-rigging, using business intelligence dashboards that combine network analytics and econometric indicators to identify suspicious patterns such as cover bidding, shared IP addresses, and structural collusion. These developments were well documented, to ensure that outputs are reproducible and facilitate rapid case analysis.
- The usage of Web Scraping to extract and download attachments from Comprasnet and PNCP, followed by metadata analysis to uncover collusive signals, such as shared document origins or authorship patterns, which proved critical in Novo Rumo's identification of coordinated bidding behaviors.
- The development of Artificial intelligence driven tools, employing unsupervised machine learning to analyze procurement documents for signs of collusion. These tools detect bidding anomalies and similarities were crucial in the Novo Rumo's case and may be useful in further analysis.
- The systematic categorization of data along two main dimensions: firm-focused (such as office locations, IP addresses, and corporate affiliations) and tender-focused (including frequent co-bidders, "rabbit" firms, and regional winners). This structured approach strengthened the evidentiary basis for identifying cartel activity and was instrumental in Novo Rumo's identification of twelve suspect firms.
- Finally, the utilization of narrative techniques to translate complex analytical findings into clear, accessible reports, making them understandable for investigators, business units, and judicial authorities. The judicial submissions produced in Novo Rumo's case exemplified this

approach, providing concise and compelling explanations of economic evidence.

Besides Novo Rumo's investigation, Cerebro's team has also developed tools for the liquid fuel retail market, an area often investigated by Cade at municipality level. Cerebro has created a specialized tool to analyze the local market for investigation, using automated filters to analyze data from the Brazilian National Agency for Petroleum, Natural Gas and Biofuels (ANP) and to identify patterns of collusion, such as price alignment or high average margin. This tool, which is currently being automated to enhance efficiency, supports investigators by prioritizing markets with potential cartel activity, streamlining case selection, and providing clear, actionable insights for both technical and judicial audiences.

As Cerebro evolves, its focus on integrating advanced analytics with user-friendly outputs positions it as a cornerstone of Cade's antitrust strategy, with successes like Novo Rumo paving the way for broader adoption of automated, data-driven tools across investigations.

III. Challenges

Cerebro's progress in leveraging Brazil's public procurement data to combat cartels is hampered by significant institutional challenges that limit its scalability and operational effectiveness. A primary obstacle is the need to simplify complex economic evidence for non-technical audiences, such as judges, to prevent misinterpretation. In the Novo Rumo operation, crafting clear, narrative-driven submissions was critical to bridging the technical-legal divide, highlighting the ongoing challenge of ensuring storytelling resonates with judicial stakeholders.

Cade's recruitment efforts face substantial hurdles due to its lack of a dedicated workforce, relying instead on seconded staff from other agencies who frequently depart for better opportunities. Securing professionals with both advanced technical skills and a nuanced understanding of Cade's antitrust objectives is a persistent challenge, as Cerebro project's dependence on specialized data analysts, which strained available resources and underscored the need for stable, skilled personnel.

Brazil's public sector buying system, marked by cumbersome bureaucratic processes, severely restricts Cerebro project's ability to acquire essential technologies, such as AI tokens and advanced analytical software, critical for scaling its data-driven capabilities. Delays in procuring these tools limited the project's ability to fully harness machine learning for real-time analysis, highlighting how protracted acquisition timelines and restrictive regulations impede Cerebro's technological advancements and investigative efficiency.

These challenges collectively underscore the need for systemic reforms to boost Cerebro project's operational capacity, from securing a stable workforce to streamlining technology procurement, ensuring the project can fully leverage Brazil's rich procurement data to drive impactful antitrust enforcement, as demonstrated by Novo Rumo's success.